

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	CH.Bhavya Sri	Reg. No.:	192411249
Section / Batch:	Slot C	Date:	22.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

A bank must evaluate loan applications to minimize default risk while maximizing profit. Analyze how algorithmic decision-making can be applied. Identify constraints such as credit score, income, and risk factors. Design a solution using optimized decision algorithms. Discuss scalability for large datasets. Evaluate computational complexity. Interpret results in financial risk management.

Code of Conduct

I CH.Bhavya Sri/192411249_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: CH.Bhavya Sri_____

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS Online Programming Lab
DAA PROGRAMMING

CHENNABOINA BHAVYA SRI
192411249RC

CO4 AT2- Industry Problem Solving Task

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Language: Python C C++ Java

QUESTIONS

Q1
Banking Loan Approval Risk Optimization (Decision Algorithms)
100 marks
1/1 answered

Q1 Banking Loan Approval Risk Optimization (Decision Algorithms) 100 marks

A bank must evaluate loan applications to minimize default risk while maximizing profit. Analyze how algorithmic decision-making can be applied. Identify constraints such as credit score, income, and risk factors. Design a solution using optimized decision algorithms. Discuss scalability for large datasets. Evaluate computational complexity. Interpret results in financial risk management.

Your Code

```
1 def evaluate_loan_applications(applicants, max_total_risk):
2     """
3     applicants: list of dicts with keys:
4         name, credit_score, income, loan amount, profit, risk_score
5     max_total_risk: maximum acceptable cumulative risk for the bank
6
7     Returns: list of approved applicants and total profit
8     """
9
10    # Step 1: Filter out applicants who fail minimum eligibility constraints
11    eligible = []
12    for app in applicants:
13        if app['credit_score'] >= 650 and app['income'] >= 25000:
```

Input

stdin input

Output

Approved Loan Applications:
C | Loan: 300000 | Profit:
25000 | Risk: 25
A | Loan: 200000 | Profit:
15000 | Risk: 20

EXPLANATION

This problem focuses on how a bank can use **decision algorithms to approve loans efficiently while reducing financial risk**. The bank receives several loan applications, and each applicant has different factors such as **credit score, income, loan amount, expected profit, and risk score**. The algorithm evaluates these factors and decides which applicants are suitable for loan approval.

First, the algorithm checks the basic **eligibility constraints**. For example, an applicant may need to have a minimum credit score and minimum income. Applicants who do not satisfy these conditions are rejected. This initial filtering reduces the number of applications that need further processing and improves the efficiency of the system.

After filtering, the algorithm considers the **profit and risk associated with each eligible applicant**. The main objective is to select applications that can provide higher profit while keeping the total risk below the bank's maximum acceptable risk limit. This can be achieved using a greedy or optimized decision-making approach that compares the profit-to-risk ratio of applicants.

The solution should also be capable of handling **large datasets**, because banks may receive thousands or millions of loan applications. Efficient algorithms are therefore important to reduce processing time. Filtering the applications requires $O(n)$ time, while sorting eligible applicants based on their profit or risk efficiency generally requires $O(n \log n)$ time.

The final output contains the **approved applicants, loan amounts, expected profits, individual risk scores, and total profit**. The bank can use these results to understand whether its lending strategy provides a good balance between profitability and risk. Thus, algorithmic loan approval helps make the decision process **faster, consistent, scalable, and data-driven**, while supporting better financial risk management.